Unmet Clinical Need Investigation

Problem Statement

Top-in-class drugs can be developed when other medications are already available for a specific condition / patient stratification, but they show significant limitations. Examples can include a limited efficacy, presence of side effects or an inconvenient dosing schedule/modality.

This investigation aims at exploring a Drug Reviews dataset[1] to find a candidate clinical condition with potential unmet clinical need, and to identify what limitations should be overcome for a novel treatment to possibly become a top-in-class.

Please note that this analysis is performed on a small set of drug reviews from 2018, so there is no claim for findings to be accurate or exhaustive at any extent.

Methods

Dataset handling

Given the no-redistribution requirement for the dataset, the source data was fetched at runtime via **ucimlrepo**, derived files were stored locally and not shared, and a locally-hosted, open-weight LLM was used for review-topic extraction and cluster labelling, and summarisation.

The dataset (4143 reviews) was initially prepared by removing reviews lacking commentary or condition information (4013 remaining), and by clustering different wordings of the same conditions (1754 distinct strings into 827 clusters). This was achieved using ***all-mpnet-base-v2*** sentence transformer and ***AgglomerativeClustering*** algorithm.

The full analysis notebook and helper functions are available at https://github.com/annadesalvo/drug-reviews.

Selection of clinical condition with unmet need

Next, conditions where at least 20 reviews were available were ranked based on a global score. This score gave equal weight to three metrics: number of available reviews, average reviews rating and percentage of reviews with a rating lower than 7. This was a simple prioritisation rule, not a validated score, and other weightings could have changed the ranking. Review ratings distribution of the 10 conditions with the lowest global scores were investigated, and the most suitable condition selected amongst these.

Unmet need investigation

An initial investigation was conducted interrogating *effectiveness* and *sideEffects* features to better understand the area of unmet need.

Competitor Drugs Analysis

The list of treatments used by reviewers was visualised.

For each treatment, percentage of reviews scoring no/marginal efficacy and percentage of reviews scoring moderate to extremely severe side effects were calculated. Percentages from drugs with more than 10 reviews were plotted in a scatter plot to understand the benchmark for a novel treatment.

Benefits and Side Effect Reviews Analysis

Different approaches were used to extract insights from the reviews.

Initially, WordCloud visualisations were employed on the subset of reviews that scored low in efficacy and on the subset of reviews that reported considerable side effects. Irrelevant words were iteratively removed manually from the visualisations.

Moving on, BERTopic was used on the full dataset of reviews for the selected condition (as clustering could become unstable on smaller subsets). Tokens and bigrams were clustered based on cosine similarity, and representations were analysed and compared with previous findings.

Next, a locally-hosted, open-weight LLM (Microsoft ***Phi-3.5-mini-instruct***, via Hugging Face Transformers, CPU inference) was employed to attempt to improve outcomes from the BERTopic-direct approach.

The LLM was initially prompted to extract all topics covered in individual reviews. Results were inspected and concatenated in a unique list, and BERTopic was used next on the embeddings list, to produce a limited number of labelled clusters.

A new LLM prompt attempted to clean and summarize the resulting representation, but it produced poor results. Instead, representations and frequencies were passed to a final prompt, where the model was asked to compile a list of the top 10 side effects.

Results

Selection of clinical condition with unmet need

The top 10 conditions with at least 20 reviews, ranked by global score are shown in Figure 1.

Condition	N_reviews	avg_rating	pct_below_7	global_score
Contraception	80	6.3	42.5	20
Sinusitis	62	6.1	41.9	24
Stop Smoking Aid	41	6.1	48.8	24
Depression	325	6.8	31.1	25
UTI	30	4.8	60.0	25
Hypercholesterolemia	74	6.4	39.2	26
Acne	216	7.1	32.4	30
Hypertension	98	7.0	31.6	32
Rosacea	27	5.7	44.4	32
Fibromyalgia	36	6.3	44.4	32

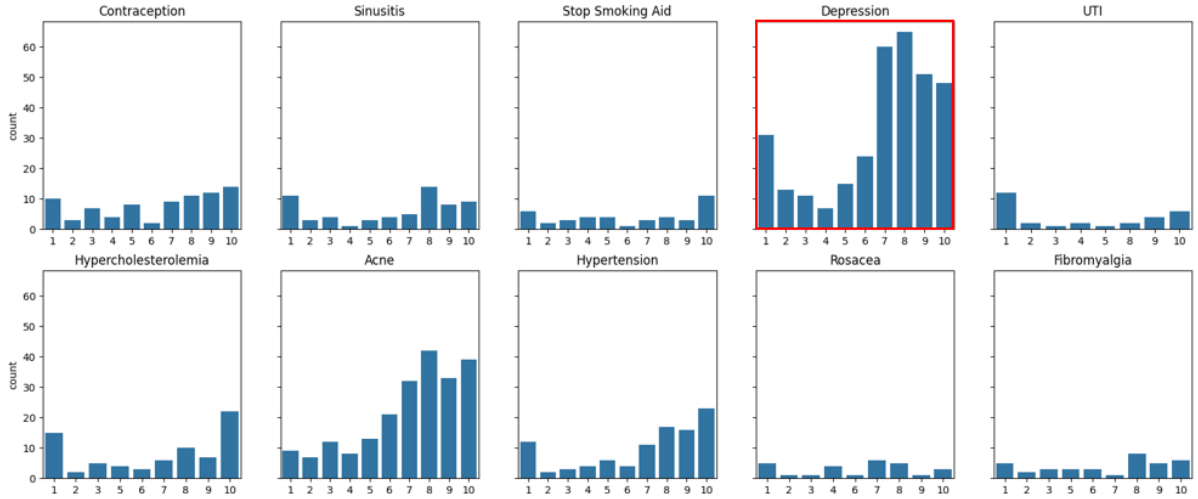


Figure 1 Top. Top-10 conditions with at least 20 reviews, ranked by global score. Bottom. Review rating distributions.

Depression was selected for further analysis, given the relatively high number of reviews (325), and for the associated rating distribution, that showed a substantial number of very poor ratings.

Unmet need investigation

Reviews were divided in three groups based on their ratings, and scores distributions for effectiveness and side effects were compared across the subgroups (results in Figure 2). Low ratings seemed to come from a mixture of low efficacy and side effects impact, giving two potential strategies for development of novel treatments.

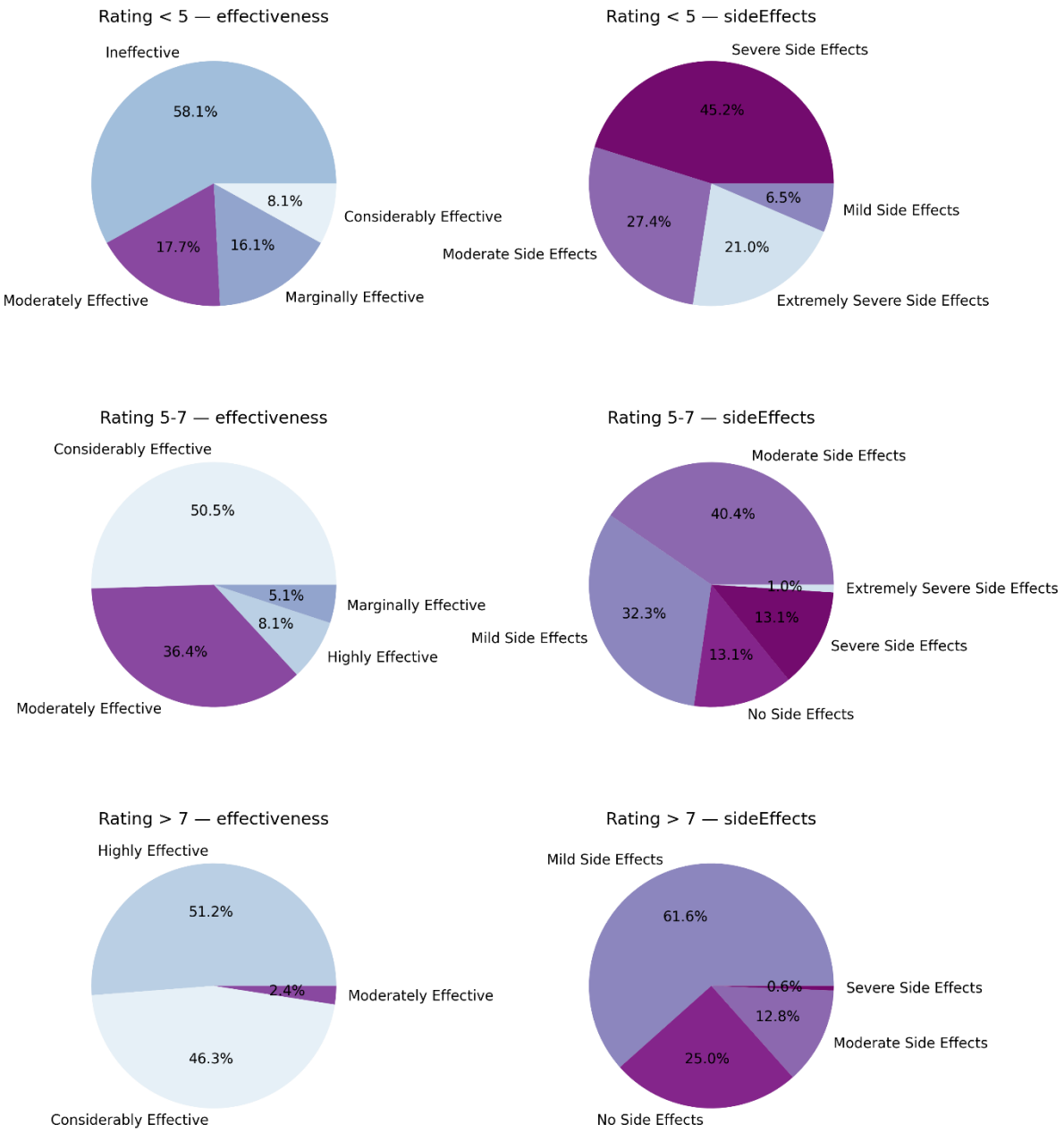


Figure 2 Effectiveness and Side effects score distributions across reviews with low / mid / high ratings

Competitor Drugs Analysis

It must be noted that this dataset doesn't include any information on patient history or on the specific treatment, so direct comparison of reviews for different drugs should not be considered as a meaningful finding without further investigations. However, the number of different treatments mentioned across the 325 reviews (33, including some brand/formulation variants) shows by itself a very crowded competitive landscape, something to be considered when evaluating new developments.

Amongst the drugs with more than 10 reviews (Figure 3), *Wellbutrin-XL* appeared to be a treatment option that had no associated negative reviews in terms of efficacy, and the lowest percentage of patients reporting moderate-to-severe side effects in the dataset (13.64%, n=22).

Should this result be confirmed in a larger, up-to-date and more complete dataset, this could become a good benchmark option for novel drug discovery campaigns.

Five treatments appeared to be associated with a higher incidence of side effects, with 50% or more patients reporting moderate-to-extreme adverse events.

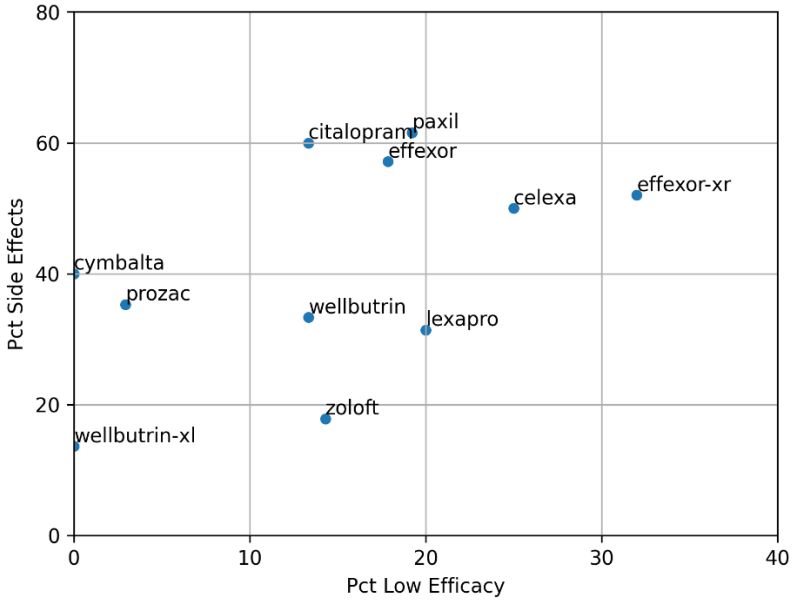


Figure 3 Percentages of reviews reporting low efficacy or considerable side effects across various treatments. Only drugs where more than 10 reviews were available are included.

Benefits and Side Effect Reviews Analysis: WordCloud approach

This analysis was performed separately on the subset of reviews scoring low efficacy and on the subset scoring remarkable side effects; please note that these sets are not independent, as the same review can appear in both. This constitutes a limitation of the specific analysis.

For both datasets, not much could be gathered from Benefits and Comments features, where single words might not be sufficient to capture the meaning of a sentence. Side effects include tokens that are clear keywords and can be sufficient to get more information.

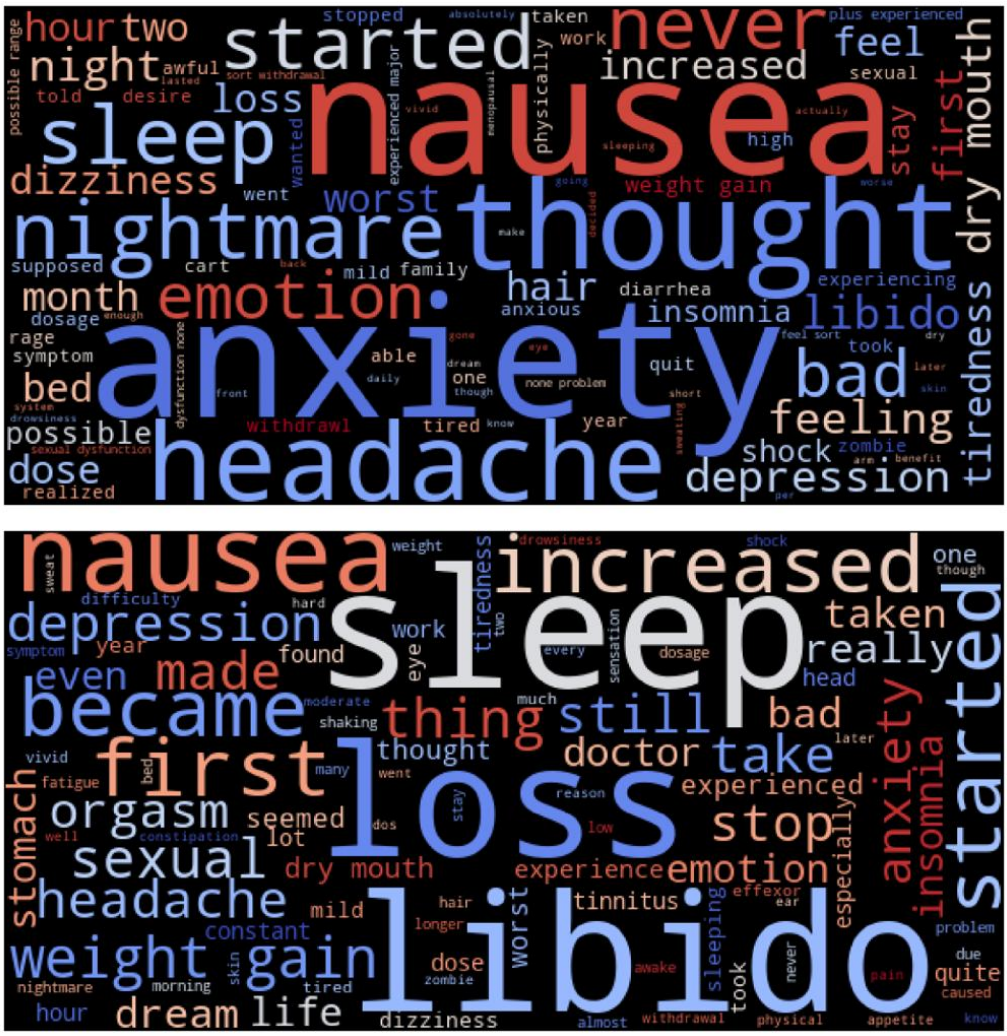


Figure 4 WordCloud visualisations for Side Effect reviews scoring low efficacy (top) and for reviews scoring moderate-to-extreme side effects (bottom)

Some common topics across the sets are already visible from the visualisations (see Figure 4), and when comparing the actual sorted tokens frequencies, recurring side effects were:

- nausea
- affected libido (sexual, libido)
- sleep disturbances (sleep)
- weight changes (weight)

Benefits and Side Effect Reviews Analysis: BERTopic-direct approach

BERTopic representations analysis on the *sideEffect* feature of the full depression review set revealed similar themes to what found in the previous analysis. Using bigrams also helped giving a better picture: *decrease sexual* / *sexual desire* / *decreased libido*, *weight loss* / *weight gain* and *elevated heart* give more robust clues than *sexual*, *weight* or *heart* alone.

Representations from the analysis included those reported in Figure 5

affected libido	0_effects_possible_loss_libido 4_sexual_decrease sexual_sexual desire_decrease
digestion issues	3_nausea_stomach_diarrhea_food
sleep disturbances and tiredness	6_drowsiness_lack_irritability_anxiety 7_insomnia_energy_did_weeks 11_patient_morning_tired_night
hyperarousal	23_aggression_sensitive_type_tempered 6_drowsiness_lack_irritability_anxiety 18_ heart_condition_constipation_elevated heart
tinnitus	9_tinnitus_severe_did experience_experience
withdrawal symptoms	14_dosage_withdrawal_physical_loss libido
dry mouth	16_dry_mouth_dry mouth_lot
weight changes	19_gain_weight gain_tired_weight 21_cravings_carbohydrates_headaches_occasional

Figure 5 Representations from BERTopic-direct approach

As visible by the representations containing multiple side effects for the same cluster, a limitation of this approach is that each review is only assigned to one cluster, despite possibly listing multiple side effects. This reduces readability of the cluster representation and causes loss in information.

The analysis on the Benefits feature was not as straightforward, as most representations didn't have clearly identifiable themes. Bigrams however were particularly useful in this case, where individual words didn't give much information: *lessened anxiety*, *improvement mood* and *noticed change* started to build a fuller picture.

Specific drug names seemed to appear in specific clusters. Possibly, similar information could be used on a larger dataset with additional metadata on patient clinical history, to rapidly identify benefits or side effects primarily associated with a specific treatment.

Benefits and Side Effect Reviews Analysis: LLM-BERTopic hybrid approach

The initial prompt, aimed at extracting themes from individual reviews, generated more than 600 unique topics, so BERTopic was used next on embeddings from the list, to produce a limited number of labelled clusters. More than 40 clusters were identified, and representations were generally more specific compared to the BERTopic-direct approach. Many side effects reconfirmed previous findings, with some appearing for the first time (see Figure 6). It can be noticed that several representations could be aggregated in broader clusters, indicating that a future improvement could be to re-tailor BERTopic parameters to encourage clustering.

withdrawal symptoms	1_withdrawal_discontinuation_lexapro_withdrawal symptoms
fatigue and sleep disturbances	3_fatigue_drowsiness_tiredness_extreme 18_trouble_sleep_difficulty_increased sleep 21_sleep_insomnia_baseline_disturbance 31_sleepiness_fatigue sleepiness_morning_fatigue 42_napping_need_insomnia_tiredness 44_energy_lowered_decrease
gastrointestinal issues	5_stomach_gastrointestinal_gastrointestinal issues_constipation 22_nausea_medication intake_sickness_mild nausea
sexual dysfunctions	6_night_skin_excessive_sexual effects
memory problems	8_memory_problems_psychological_loss 26_alteration_disturbances_insomnia_memory
weight changes	9_weight_weight loss_weight gain_gain 14_appetite_loss appetite_change weight_importance 23_body_changes_change_head
hyperarousal	10_irritability_inhibition_extreme_agitation 13_cardiovascular_racing_function_increase 20_mood_mood swings_swings_changes 28_electric_shakiness_racing_jitteriness 30_anxiety_attacks_panic attacks_nervousness
headaches	24_migraines_shakiness__ 32_headaches_migraines_frequency_food
cramps	33_cramps_pain_leg_severe
emotional dulling	34_emotional_dulling_emotional numbness_numbness
dry mouth	35_dry_mouth_dry mouth_sleepiness lack

Figure 6 Representations from BERTopic applied to a list of LLM-extracted side effects

Finally, when the LLM was given the representations list together with their frequencies, and was asked to compile a list of the top 10 side effects, this was the output:

- *decreased sexual performance*
- *withdrawal symptoms*
- *fatigue*
- *stomach issues*
- *memory loss*
- *weight change*
- *irritability*
- *treatment efficacy* (this is likely coming from noisy clusters)
- *appetite loss*
- *sleep disturbances*

Overall, using LLMs performed well to extract topics from the individual reviews, but looking at findings the gain was marginal, considering the computational cost of analysing the individual reviews. Results were cleaner, but the top side effects were already captured with the simpler methods. The limitation came from the necessity of using a local lightweight model, as the dataset could not be shared to third parties, and the LLM often disregarded formatting instructions, adding noise to the otherwise cleaner results.

Conclusions

Amongst the various conditions, depression was identified as potentially carrying unmet clinical need. This was due to the relatively higher percentage of poorer scores in associated reviews, although the selection was limited by the small dataset, since conditions with more reviews had to be prioritized.

Low scores were linked both to limited efficacy and to considerable side effects, potentially giving multiple approaches for future developments.

The multitude of treatments mentioned across the reviews depicts a crowded competitive landscape, and from the available reviews there seem to be treatments that offer good efficacy and low side effects. Given the limited available information, however, no direct comparisons can be made between treatments, and subsequent analysis was performed across all reviews, regardless of the reviewed drug.

Focusing on side effect reviews, similar findings were corroborated using different approaches, and some recurring events across the methods included:

- *decreased sexual performance*
- *withdrawal symptoms*
- *fatigue and sleep disturbances*
- *gastrointestinal issues*
- *hyperarousal*
- *dry mouth*
- *headaches*

Many of these symptoms are difficult to measure preclinically, so mechanistic drivers should be investigated to properly counter-screen in early development. Feasibility of this should be carefully considered.

Please note that this analysis was performed on a small set of drug reviews from 2018, so there is no claim for findings to be accurate or exhaustive at any extent.

Bibliography

1. Kallumadi, S. & Gräßer, F. (2018). Drug Reviews (Druglib.com) [Dataset]. UCI Machine Learning Repository. https://doi.org/10.24432/C55G6J. CC BY 4.0 with research-use-only, non-commercial, no-redistribution conditions.